Assessment Exercise: Mini AI Gateway

Duration: 180 minutes (individual)

Starting point: your RAG lab code (Basic RAG, MQR, MQR + RRF, Agentic RAG)

Language: Python

Week - 06

1. Problem statement

In our RAG application, we send the user's query and retrieved passages directly to the LLM to generate an answer. If the query or any passage contains a secret, that secret is sent to the model.

Build a **Mini AI Gateway**: a layer that sits just before the LLM call and checks the query and the passages for three types of secret.

- If a **passage** contains a secret, drop that passage. Send the remaining passages to the LLM.
- If the **query** contains a secret, block the request. Do not call the LLM.

Build it so that new secret types can be added later without changing the gateway's core logic. You will also explain your design approach.

```
query -> retriever -> passages -> [ Mini AI Gateway ] -> LLM -> answer
                                          |
                                          +-> block request/drop passage
```

2. Secret types to cover

Secret type	Examples (synthetic)	Detection rule
Credentials and API keys	`sk-test-9f8e7d6c5b4a3f2e1d0c`, `AKIA4EXAMPLE7KEY9XYZ`, `password=Winter2026!`, `client_secret: Qx7abc`	Known key prefixes, and the words password, secret or api_key followed by = or :
Payment card numbers	`4111 1111 1111 1111`, `4111-1111-1111-1111`	13 to 19 digits (spaces or dashes allowed) that pass the Luhn check
Confidential deal codenames	Project Falcon, Project Monsoon	Case-insensitive match against a list kept in a config file

Use only synthetic data. Do not use real client or colleague data.

3. Expected behaviour

Situation	Result
No secret found	Normal answer
Secret in one or more passages	Those passages are dropped; the LLM answers from the rest
Secret in every passage	No LLM call. Reply: "I could not find this in the approved sources."
Secret in the query	No LLM call. Reply: "Your request contains restricted content and was not processed."
Error inside the gateway	No LLM call (fail closed)

Minimum interface (you may add more):

```
def check(query: str, passages: list[Passage]) -> GatewayResult

GatewayResult:
    decision        "ALLOW" | "BLOCK_QUERY" | "BLOCK_CONTEXT" | "BLOCK_ERROR"
    safe_passages   passages allowed through
    dropped         chunk_ids that were dropped
    findings        count per secret type, e.g. {"CREDENTIAL": 1, "CODENAME": 2}
```

4. Constraints

- Never write the secret text to logs, error messages or API responses. Log only the secret type and counts.
- The check should take under 20 ms for one query plus 5 passages of about 2,000 characters.

5. Deliverables

- Submit the Judge Test report
- Design approach discussion with Trainer